

The excessive costs of transacting has ushered in many fintech innovations. PayPal,<sup>3</sup> founded more than 20 years ago, is a forerunner in the payments space; in 2017, seven of the largest U.S. banks added their own payment system called Zelle.<sup>4</sup> An important commonality of these cost-reducing fintech advances is that they rely on the centralized backbone of the current financial infrastructure.

## **BITCOIN AND CRYPTOCURRENCY**

The dozens of digital currency initiatives beginning in the early 1980s all failed.<sup>5</sup> The landscape shifted, however, with the publication of the famous Satoshi Nakamoto Bitcoin white paper<sup>6</sup> in 2008, which presents a peer-to-peer system that is decentralized and uses the concept of *blockchain*. Invented in 1991 by Haber and Stornetta,<sup>7</sup> blockchain was initially primarily envisioned to be a time-stamping system to keep track of different versions of a document. The key innovation of Bitcoin was to combine the idea of blockchain (time stamping) with a consensus mechanism called *proof of work* (introduced by Back<sup>8</sup> in 2002). The technology produced an immutable ledger that eliminated a key problem with any digital asset: you can make perfect copies and spend them multiple times. Blockchains allow for the important features desirable in a store of value, which were never before simultaneously present in a single asset. Blockchains allow for cryptographic scarcity (Bitcoin has a fixed supply cap of 21 million), censorship resistance and user sovereignty